NPTEL ONLINE CERTIFICATION COURSE

Introduction to Aerospace Engineering - Flight

Prof. Rajkumar S. Pant, IIT Bombay

COURSE ID **101101079**	DURATION **12 weeks**
CREDITS **4**	LEVEL **Undergraduate**
TYPE **Core**	LANGUAGE **English**

ABOUT THE COURSE

The aim of this course is to provide a general overview of the field of Aeronautical Engineering to interested students. The course will consist of ten Capsules, each consisting of two Lectures. Each Lecture will cover a specific concept or area relevant to the subject. An attempt will be made to cover the contents in an interesting manner, by a judicious use of a mix of powerpoint presentations, in-class activities, quizzes, innovative and hands on assignments that will not only increase the awareness of the students, but also satiate their curiosity and desire to know more about the various concepts related to the subject.

INTENDED AUDIENCE : Any discipline of Engineering, who are interested in the subject

INDUSTRY SUPPORT : DRDO, HAL, NAL, IAF

COURSE DETAILS

Category

AERO_COURSES, Domain_22

COURSE LAYOUT

WEEK	TOPICS
Week 1	Atmosphere and its properties
Week 2	Nomenclature of aircraft components
Week 3	Fluid Mechanics – I : Incompressible flow, Bernoulli's Equation, Coanda Effect, and Mach No..
Week 4	Fluid Mechanics –II : Viscous Flow, Boundary Layer, Pressure Measurement
Week 5	Aerodynamics – I : Airfoils, and Lift Generation Theories
Week 6	Aerodynamics – II : Critical Mach no., Types of Drag
Week 7	Propulsion : Types of Aircraft Engines
Week 8	Aircraft Performance - I : Steady Level Flight and Altitude effects
Week 9	Aircraft Performance- II : Glide, Climb, Ceilings, Turn, and Pull up
Week 10	Aircraft Longitudinal Stability,and V-n Diagram
Week 11	Aircraft Performance- III : Takeoff and Landing, Range and Endurance, Range-Payload Diagram
Week 12	Flapping Wing Aerodynamics

BOOKS AND REFERENCES

Anderson, J. D., The Aeroplane, a History of its Technology, AIAA Education Series, 2002
Eberhardt S, Anderson D., Understanding Flight. McGraw-Hill Publishing; 2009.
Anderson, J. D., Introduction to Flight, McGraw-Hill Professional, 2005
Ojha S.K., Flight Performance of Aircraft, AIAA Education Series, 1995 Existing video clips of this course recorded by CDEEP in 2017 to be used with some editing.

INSTRUCTOR BIO

Prof. Rajkumar S. Pant
IIT Bombay

Prof. Rajkumar S. Pant has Bachelors, Masters and Ph.D. degrees in Aerospace Engineering. His areas of specialization include Aircraft Conceptual Design, Air Transportation, and Optimization. He has been a member of faculty of Aerospace Engineering Department at the Indian Institute of Technology Bombay since December 1989. Prof. Pant is an alumnus of College of Aeronautics, Cranfield University, UK, where he earned his Ph.D. under Commonwealth Scholarship Scheme, IIT Madras, where he did his Masters in Aeronautical Engineering, and PEC Chandigarh where he underwent his undergraduate studies in Aeronautical Engineering. He has also worked for five years in Hindustan Aeronautics Limited in the Design & Engineering Department at Kanpur (3.5 years) and Nasik (1.5 years) Divisions. He has published and presented ~ 220 scientific papers, of which ~ 170 are in international journals and conferences. He has also visited several top ranking institutes and universities all over the world. Prof. Pant was a Visiting Professor at School of Mechanical & Aerospace Engineering at Nanyang Technological University, Singapore in 2015-16, visiting faculty at Department of Aerospace & Ocean Engineering at Virginia Polytechnic Institute and State University in 2010-11, and a visiting researcher at Instituto Tecnolgico de Aeronutica, Brazil in 2012, Texas A&M University in 2011, Cambridge University in 2008, and Imperial College London in 2006. In November 2012, he was appointed as a Special Visiting Researcher under the Science Without Borders program of the Brazilian Government for three years

CERTIFICATION

The course is free to enroll and learn from. But if you want a certificate, you have to register and write the proctored exam conducted by us in person at any of the designated exam centres.

The exam is optional for a fee of Rs 1000/- (Rupees one thousand only).

Date and Time of Exams: **October 18, 2026** Morning session 9am to 12 noon; Afternoon Session 2pm to 5pm.

Registration url: Announcements will be made when the registration form is open for registrations.

The online registration form has to be filled and the certification exam fee needs to be paid. More details will be made available when the exam registration form is published. If there are any changes, it will be mentioned then.

Please check the form for more details on the cities where the exams will be held, the conditions you agree to when you fill the form etc.

Certificate will have your name, photograph and the score in the final exam with the breakup.It will have the logos of NPTEL and IIT Bombay .It will be e-verifiable at nptel.ac.in/noc.

Only the e-certificate will be made available. Hard copies will not be dispatched.

Once again, thanks for your interest in our online courses and certification. Happy learning.

- NPTEL team